

SAFETY-GATED SPECTRAL RECONSTRUCTION OF EXPERIMENTAL RAMSEY FRINGES

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Abstract. A single reconstruction rule is often poorly matched to a heterogeneous collection of experimental fringes. A trace-local procedure is introduced that retains linear interpolation unless nested development points support a regularized spectral model by a fixed safety margin. Each spectral candidate uses 4, 6, 8, or 10 harmonics and a Bohr-anchored Sobolev penalty; independent trace blocks are solved without materializing the global block-diagonal design. A deterministic selection theorem bounds the selected test loss in terms of the best candidate whenever development and test losses are uniformly close. A second theorem proves exact equivalence of block and explicit normal equations and reduces design storage from $O(NTp)$ to $O(Np)$. Evaluation uses 40,787 measured probabilities from 177 trapped-ion Ramsey fringes in an authenticated CC-BY archive. Every fourth phase point is held out; disjoint outer-training points select among 29 candidates per trace. The fixed five-percent gate lowers mean held-out RMSE from 0.09434 to 0.09146 and pooled RMSE from 0.09947 to 0.09732. Improvement occurs in every state-group mean and 69 individual traces; unfavorable accepted-model cases are retained. At 32 traces, the equivalent block solve is 17.1 times faster in the recorded CPU experiment and stores 1.03 MB rather than 31.55 MB. These results establish a guarded reconstruction and representation improvement on the stated archive, not universal spectral superiority or dynamical-system identification.

1. Measured-fringe setting. The record contains $N = 40,787$ excitation probabilities from $T = 177$ Ramsey fringes spanning eight prepared-state labels [1]. Numeric worksheet cells are transcribed without loss to a checksum-locked CSV. The complete archive and derived-file SHA-256 digests are stored in the released result record. Every reported result is regenerated from that file.

For trace t , measurements (ϕ_{ti}, y_{ti}) are divided by phase index. Indices congruent to zero modulo four form the outer test set. Among the remaining points, indices congruent to three modulo eight form an inner development set; all other points form the inner fit set. Thus no outer-test probability participates in candidate construction or selection.

Piecewise-linear interpolation is a strong local comparator for these densely sampled fringes. A global ten-harmonic fit improves the cleanest state group but degrades the heterogeneous collection. The proposed rule treats linear interpolation as a protected default and permits a spectral replacement only when its inner-development improvement exceeds a predeclared margin.

2. Safety-gated block spectral fit. For a candidate harmonic count h , define

$$\begin{aligned} f_h(\phi) &= [1, \sin \phi, \cos \phi, \dots, \\ &\quad \sin(h\phi), \cos(h\phi)] \in \mathbb{R}^p, \\ p &= 2h + 1. \end{aligned} \tag{2.1}$$

Let F_t contain inner-fit rows and y_t the corresponding probabilities. Candidate coefficients solve

$$\hat{\theta}_t(h, \lambda) = \arg \min_{\theta} \|F_t \theta - y_t\|_2^2 + \theta^T (10^{-2} I + \lambda D_h) \theta, \tag{2.2}$$

where the intercept and fourth harmonic have zero entries in D_h , while the sine and cosine entries for harmonic $k \neq 4$ equal k^2 . The grid is $h \in \{4, 6, 8, 10\}$ and $\lambda \in \{0, 0.03, 0.1, 0.3, 1, 3, 10\}$.

Let $\hat{L}_0$ be linear-interpolation RMSE on inner development points and $\hat{L}_s$ the minimum spectral-development RMSE. With $\tau = 0.05$, the spectral minimizer is accepted only if

$$\hat{L}_s < (1 - \tau) \hat{L}_0. \tag{2.3}$$

The accepted model is refitted on all outer-training points. Otherwise linear interpolation is retained. This gives 29 candidates, including the protected default, for every trace.

Theorem 1 (Conditional selection bound). *Let $\mathcal{C}$ contain the default and all spectral candidates. Suppose development and test losses obey $|\hat{L}_c - L_c| \leq \epsilon$ for every $c \in \mathcal{C}$. The rule in Eq. (2.3) returns $\hat{c}$ with*

$$L_{\hat{c}} \leq \frac{\min_{c \in \mathcal{C}} L_c + \epsilon}{1 - \tau} + \epsilon. \tag{2.4}$$

Table 1: Outer-test RMSE by prepared-state label. Values are means over traces; the last column records accepted spectral models.

State	Traces	Gated	Linear	Selected
0 + 1	11	.04733	.05484	9
0 + 2	10	.04780	.05191	5
0 + 3	48	.10693	.10772	15
0 + 4	41	.11246	.11454	18
0 + 6	24	.12774	.13086	16
1 + 2	15	.05646	.05984	8
1 + 3	12	.06156	.06436	8
2 + 3	16	.04973	.05613	12
All	177	.09146	.09434	91

Proof. If the spectral candidate is accepted, it is the global development minimizer. If the default is retained, the best spectral loss is at least $(1 - \tau)\hat{L}_0$, hence $\hat{L}_0 \leq \min_c \hat{L}_c / (1 - \tau)$. Both cases give $\hat{L}_c \leq \min_c \hat{L}_c / (1 - \tau)$. Applying the assumed discrepancy to the selected candidate and to the test-loss minimizer proves Eq. (2.4). $\square$

The theorem is deterministic and conditional. It exposes the role of the safety margin without asserting that one inner slice estimates future loss uniformly. The trace-level confirmation panel reports observed failures of that premise.

3. Matrix-free block elimination. Let $F = \text{diag}(F_1, \dots, F_T)$ and $R = \text{diag}(R_1, \dots, R_T)$ denote the explicit global design and regularizer. The implementation stores local feature rows and trace indices, applying F and F^T by indexed reductions.

Theorem 2 (Exact block equivalence and cost). *If every $F_t^T F_t + R_t$ is positive definite, the explicit solution equals the concatenation of*

$$\hat{\theta}_t = (F_t^T F_t + R_t)^{-1} F_t^T y_t. \quad (3.1)$$

For at most p local features and N total fit rows, block elimination costs $O(Np^2 + Tp^3)$ arithmetic and $O(Np + Tp^2)$ storage. Explicit global design storage is $O(NTp)$.

Proof. The normal matrix is

$$F^T F + R = \text{diag}(F_1^T F_1 + R_1, \dots, F_T^T F_T + R_T). \quad (3.2)$$

Its inverse contains the local inverses, proving equality. Accumulating local Gram matrices costs $O(Np^2)$, their factorizations cost $O(Tp^3)$, and local features plus Gram matrices require $O(Np + Tp^2)$ storage. An explicit row has Tp columns, giving $O(NTp)$ storage. $\square$

4. Experimental results. The gate accepts a spectral model for 91 of 177 traces. Table 1 shows that mean held-out RMSE decreases in every state group. Across all traces, mean RMSE falls by 3.04 percent and pooled squared error falls by 4.28 percent.

Figure 1(a) displays the state-stratified comparison. Panel (b) retains every trace rather than only a favorable subset. The gate improves 69 traces, ties 86 traces that retain interpolation, and loses on 22 accepted spectral traces. Panel (c) relates the inner selection statistic to outer-test gain; points below zero are explicit selection failures.

The representation comparison in Table 2 solves identical normal equations and verifies predictions to 9.44×10^{-16} at 32 traces. The full 177-trace fit stores 5.37 MB in local features and indices; the corresponding explicit design would occupy 0.907 GB. These numbers refer to representation and solve cost, while nested selection performs additional small local solves.

5. Scope. The archive supplies broad within-experiment heterogeneity but not independent devices or acquisition campaigns. The five-percent gate is fixed before the outer evaluation, yet the conditional discrepancy in Theorem 1 is not certified from one inner slice. The 22 accepted-model losses remain essential evidence. Spectral coefficients describe measured phase fringes and do not identify

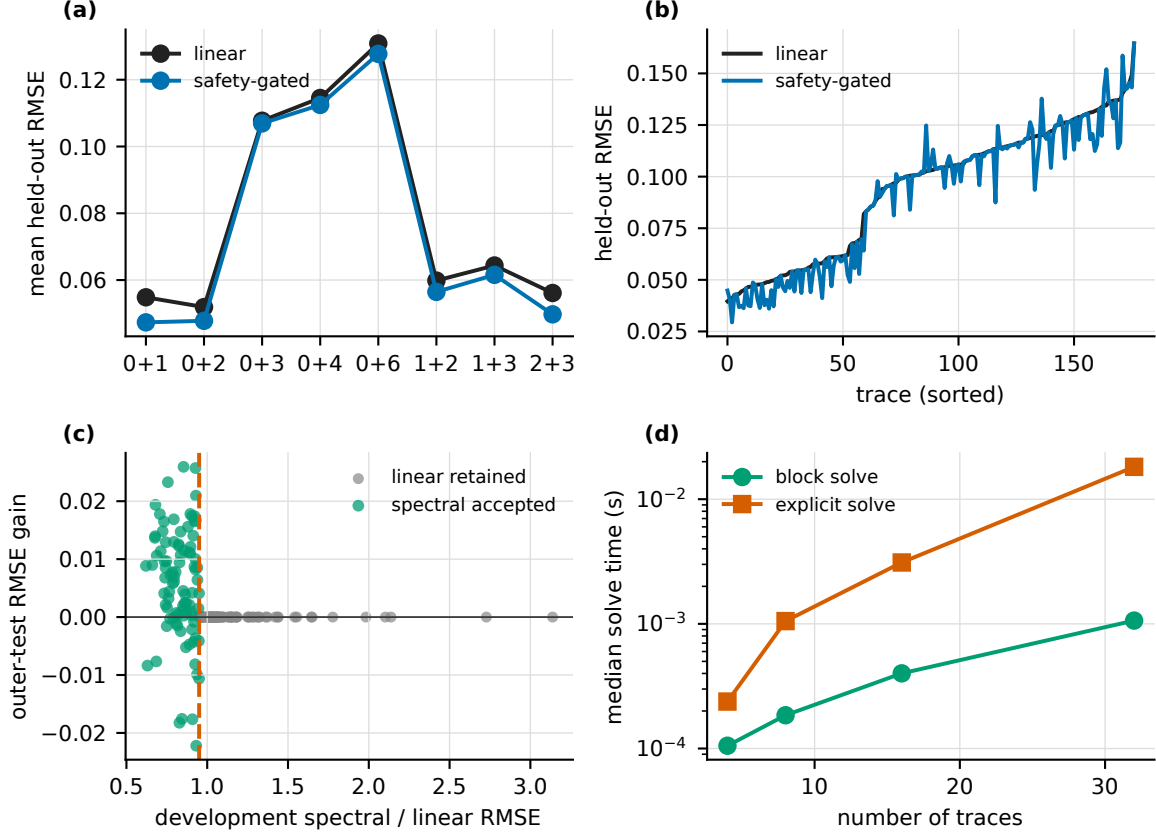


Fig. 1: Accuracy, selection, and cost evidence. (a) Mean outer-test RMSE by state group. (b) Every trace, sorted by interpolation RMSE. (c) Inner spectral-to-linear error ratio against outer-test gain; the dashed line is the fixed gate. (d) Equivalent explicit and block-solve CPU time.

a Hamiltonian or open-system generator. The supported conclusion is computational and predictive: guarded local selection improves aggregate held-out reconstruction on this archive while preserving an interpolation fallback and an exact block-elimination implementation.

References.

- [1] A. Kovalenko et al. Experimental data—quantum non-gaussian coherences of an oscillating atom, 2025.

Table 2: Measured equivalent-solve scaling. Times are medians of three CPU runs; memory is stored design representation.

Traces	Block (ms)	Explicit (ms)	Speedup	MB ratio
4	0.105	0.238	2.3	3.8
8	0.185	1.052	5.7	7.6
16	0.401	3.108	7.8	15.3
32	1.062	18.181	17.1	30.5